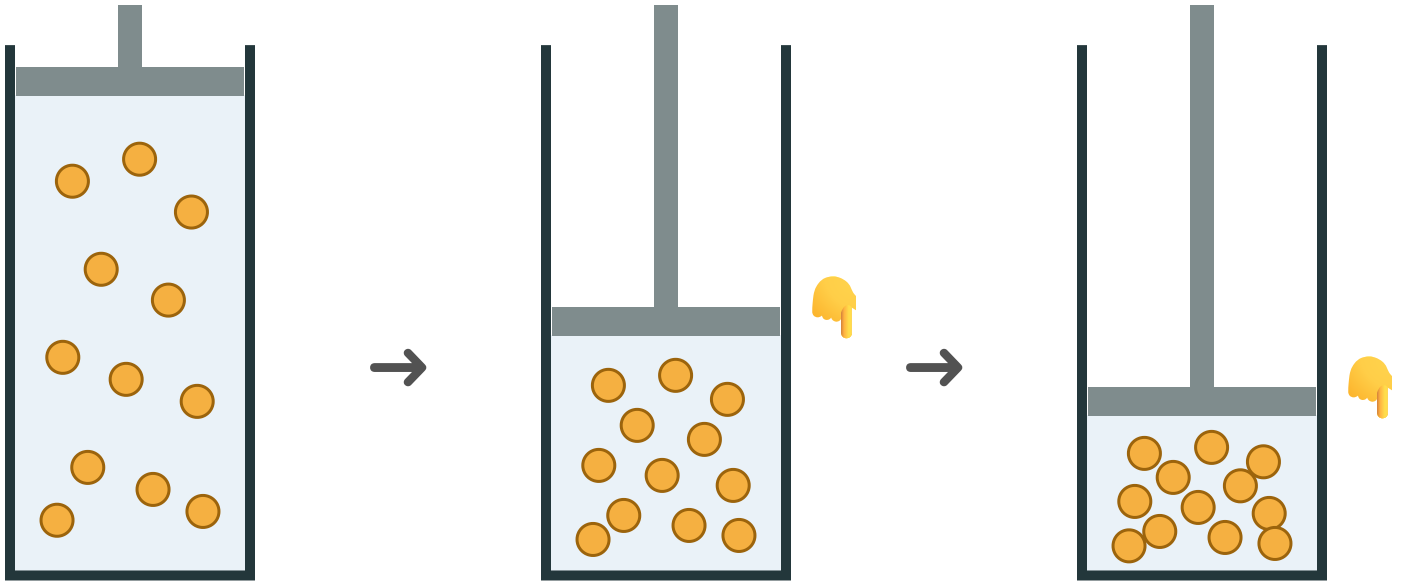
Boyle-Mariotte's law: the temperature does not change



V = 6 L
p = 1 atm

V = 3 L
p = 2 atm

V = 2 L
p = 3 atm

T constant

$$p_1 \cdot V_1 = p_2 \cdot V_2$$

p (atm)	V (L)	p · V
1	6	**6**
2	3	**6**
3	2	**6**

Same particles in **less space** hit the walls **more often**: half the volume → double the pressure. The product **p · V** stays the same.